

22CH203- ENGINEERING CHEMISTRY – II

UNIT-II- Energy Storing Devices

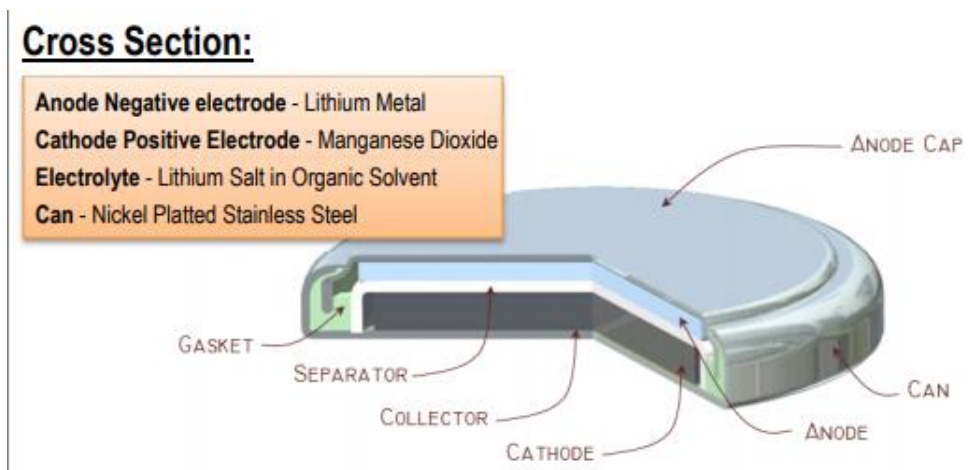
Topic 2.5: Construction, working and applications: Lithium batteries (Li-MnO₂ Batteries and Lithium ion battery)

1. Lithium battery

Lithium Batteries are disposable (primary) batteries that have lithium metal or lithium compounds as an anode. Depending on the design and chemical compounds used, lithium cells can produce voltages from 1.5 V (comparable to a zinc-carbon or alkaline battery) to about 3.7 V. By comparison, lithium-ion batteries are rechargeable batteries in which lithium ions move between the anode and the cathode, using an intercalated lithium compound as the electrode material instead of the metallic lithium used in lithium batteries.

1.1 Li-MnO₂ Batteries: (Lithium metal battery)

- Li-MnO₂ cell is one of the most widely used batteries
- It consists of Li anode and MnO₂ as cathode.
- Lithium halide in an organic solvent solution is used as electrolyte.
- The MnO₂ pellet and anode disc are separated by polypropylene separator pasted with electrolyte. The cell is sealed with can serving as the cathode terminal and cap as the anode terminal.



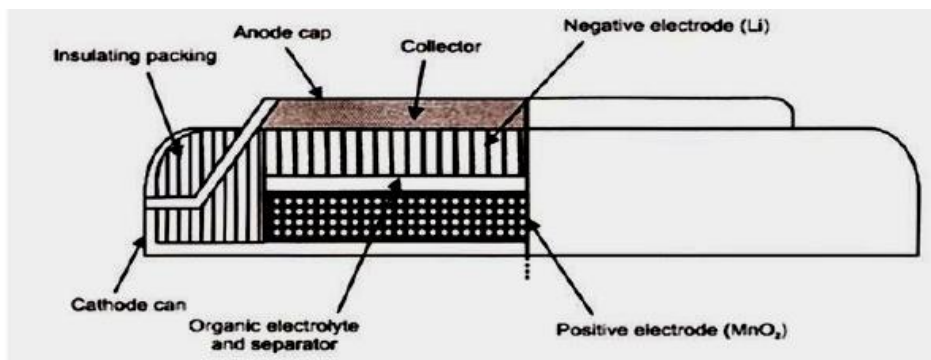
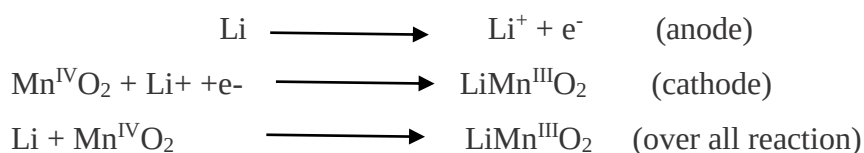


Fig. Li-MnO₂ battery



In the above cell reaction Mn reduces from +4 oxidation state to +3 oxidation state. Lithium ion enters into the MnO₂ crystal lattice.

The voltage of the cell is upto 4V depending on the cathode used.

Uses:

It is mainly used in applications watches, calculators, cameras, lighting equipment sand long time memory backups.

Advantages:

- High current density due to low atomic mass of lithium Operates between wide range of temperature (-40 to 70⁰C)
- Low self-discharge rate and shelf life is high

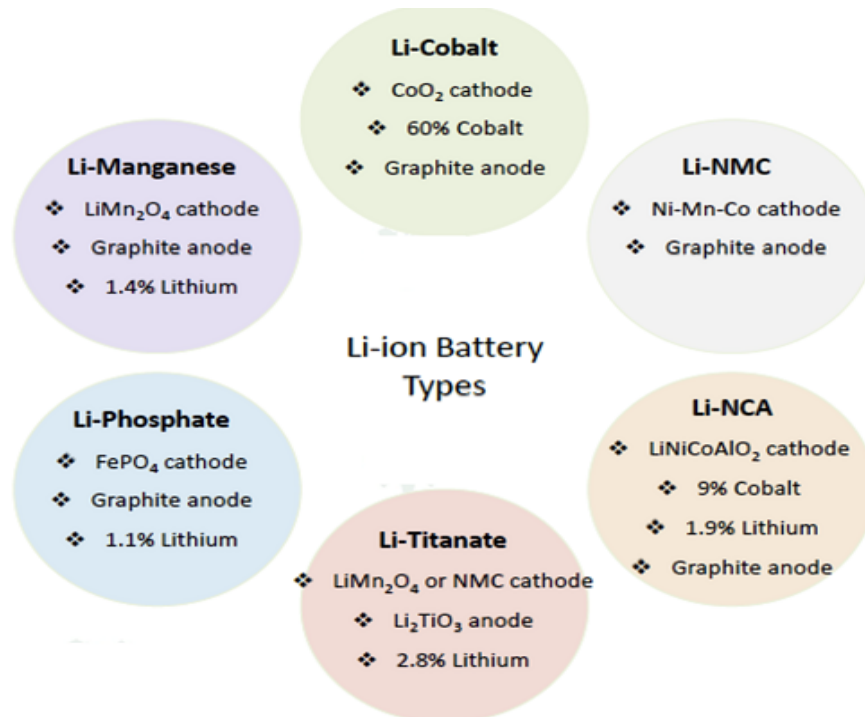
Small lithium batteries are very commonly used in small, portable electronic devices, such as PDAs, watches, camcorders, digital cameras, thermometers, calculators, personal computer BIOS (firmware), communication equipment and remote car locks. They are available in many shapes and sizes, with a common variety being the 3 volt "coin" type manganese variety, typically 20 mm in diameter and 1.6–4 mm thick.

The heavy electrical demands of many of these devices make lithium batteries a particularly attractive option. In particular, lithium batteries can easily support the brief, heavy current

demands of devices such as digital cameras, and they maintain a higher voltage for a longer period than alkaline cells.

Lithium-ion Battery (Cathode) Types and Usage Areas

- Lithium-ion lithium cobalt oxide battery (ICR)
- Lithium–silicon battery.
- Lithium-ion manganese-oxide battery (LMO)
- Lithium-ion polymer battery (LiPo)
- Lithium–iron–phosphate battery (LFP)
- Lithium–nickel–manganese–cobalt oxides (NMC)
- Lithium–nickel–cobalt–aluminium oxides (NCA)
- Lithium–sulfur battery.



1.2 Lithium-ion battery (Li-ion Battery)

- Li-ion batteries are secondary batteries.
- The battery consists of an anode of Lithium, dissolved as ions, into a carbon.
- The cathode material is made up from Lithium liberating compounds, typically the three electro-active oxide materials,
 - *Lithium Cobalt-oxide (LiCoO_2)*

- *Lithium Manganese-oxide (LiMn_2O_4)*
- *Lithium Nickel-oxide (LiNiO_2)*

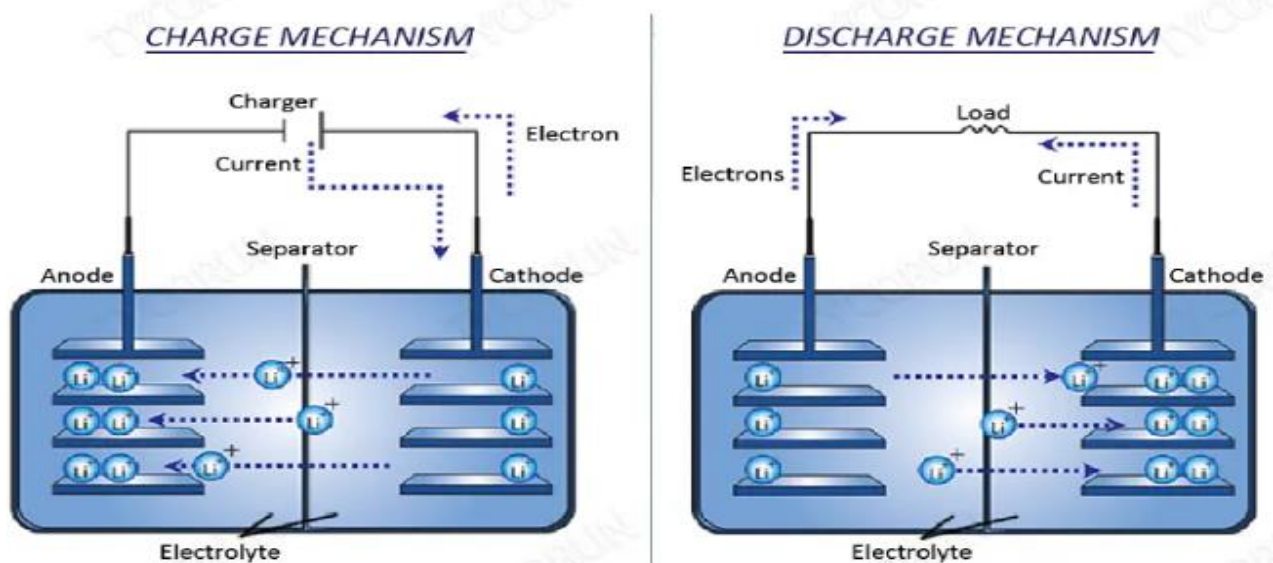
Principle

- During the charge and discharge processes, lithium ions are inserted or extracted from interstitial space between atomic layers within the active material of the battery.
- Li-ion is transfers between anode and cathode through lithium electrolyte. Since neither the anode nor the cathode materials essentially change, the operation is safer than that of a Lithium metal battery.

Construction

- Li-ion cell has a four-layer structure.
- A positive electrode made with Lithium Cobalt Oxide has a current collector made of thin aluminum foil - cathode.
- A negative electrode made with specialty carbon has a current collector of thin copper foil - anode.
- A separator is a fine porous polymer film.
- An electrolyte made with lithium salt in an organic solvent.
- The electrolytes are selected in such a way that there should be an effective transport of Li-ion to the cathode during discharge.
- The type of conductivity of electrolyte is ionic in nature rather than electronic

Working:



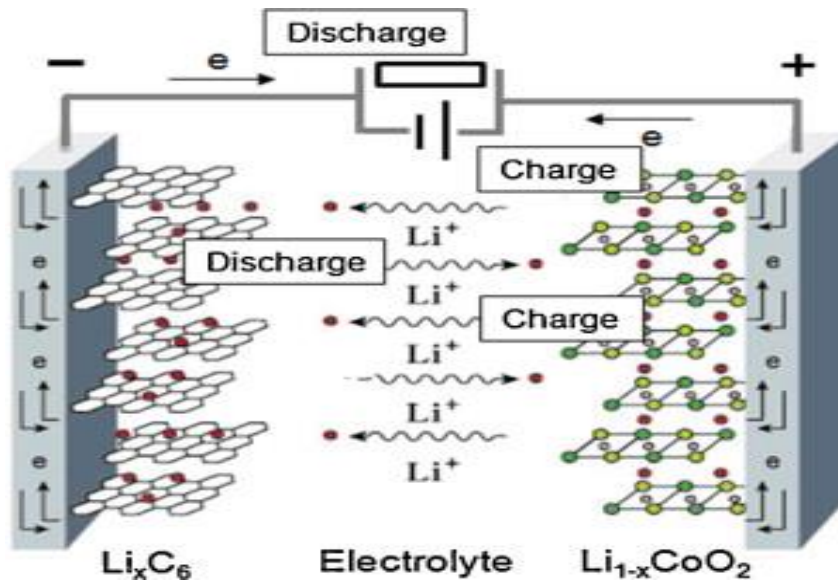


Fig. Working of lithium ion battery

- The traditional batteries are based on galvanic action but Lithium ion secondary battery depends on an "*intercalation*" mechanism.
- This involves the insertion of lithium ions into the crystalline lattice of the host electrode without changing its crystal structure.
- These electrodes have two key properties. One is the open crystal structure, which allow the insertion or extraction of lithium ions and the second is the ability to accept compensating electrons at the same time. Such electrodes are called intercalation hosts.
- The chemical reaction that takes place inside the battery is as follows, during charge and discharge operation:
- The lithium ion is inserted and exerted into the lattice structure of anode and cathode during charging and discharging
- During discharge current flows through external circuit and light glows
- During charging, no the electrons flows in the opposite direction.
- During charging, lithium in positive electrode material is ionized and moves from layer to layer and inserted into the negative electrode.
- During discharge Li ions are dissociated from the anode and migrate across the electrolyte and are inserted into the crystal structure of the host compound of cathode.

- *Role of electrolyte is to act as a medium for the transfer of ions between the two electrodes.*
In general, lithium salt dissolved in organic solvent is used as electrolyte in lithium ion batteries.
- At the same time the compensating electrons travel in the external circuit and are accepted by the host to balance the reaction.
- The process is completely reversible. Thus the lithium ions pass back and forth between the electrodes during charging and discharging.

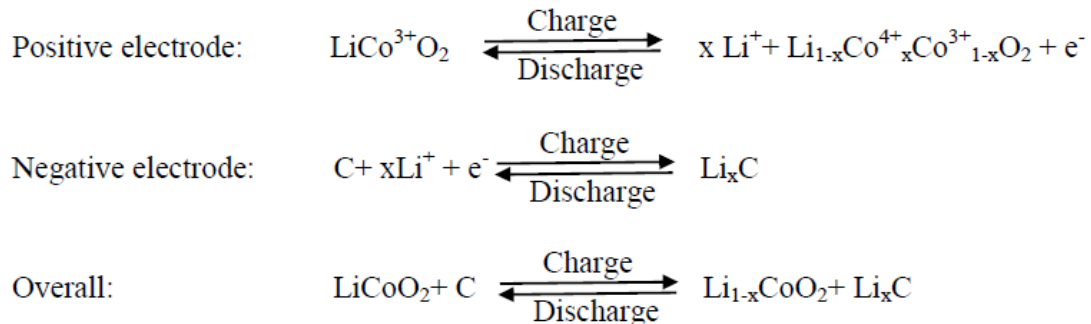


Figure: Schematics of Li-ion battery using LiCoO₂ as cathode and graphite as anode

Advantages

- They have high energy density than other rechargeable batteries.
- They are less weight.
- They produce high voltage out about 4 V as compared with other batteries.
- They have improved safety, i.e. more resistance to overcharge.
- No liquid electrolyte means they are immune from leaking.
- Fast charge and discharge rate.
- Lithium-ion batteries have higher energy densities than lead-acid batteries or nickel-metal hydride batteries, so it is possible to make the battery size smaller than others while retaining the same storage capacity.

Disadvantage

- They are expensive.
- They are not available in standard cell types.

Applications

- The Li-ion batteries are used in cameras, calculators.
- They are used in cardiac pacemakers and other implantable device.
- They are used in telecommunication equipment, instruments, portable radios and TVs, pagers.
- They are used to operate laptop computers and mobile phones and aerospace application.

Summary:

Type	Description
Lithium cobalt oxide (LiCoO_2)	The battery consists of a cobalt oxide cathode and a graphite carbon anode. The cathode has a layered structure. During discharge, lithium ions move from the anode to the cathode. The flow reverses on charge. The drawback of Li-cobalt batteries is their relatively short life span, low thermal stability, and limited load capabilities.
Lithium manganese oxide (LiMn_2O_4)	Li-ion with manganese spinel was first published in the <i>Materials Research Bulletin</i> in 1983. The architecture forms a three-dimensional spinel structure that improves ion flow on the electrode, resulting in lower internal resistance and improved current handling. A further advantage of the spinel structure is high thermal stability and enhanced safety, but the cycle and calendar life are limited.
Lithium nickel manganese cobalt oxide (LiNiMnCoO_2 , or NMC)	One of the most successful Li-ion systems is a cathode combination of nickel-manganese-cobalt (NMC). Similar to Li-manganese, these systems can be tailored to serve as energy cells or power cells.
Lithium iron phosphate (LiFePO_4)	In 1996, the University of Texas (and other contributors) discovered phosphate as cathode material for rechargeable lithium batteries. Li-phosphate offers good electrochemical performance with low resistance. This is made possible with nanoscale phosphate cathode material. The key benefits are high current rating and long cycle life, besides good thermal stability, enhanced safety, and tolerance if abused.
Lithium titanate ($\text{Li}_4\text{Ti}_5\text{O}_{12}$)	Batteries with lithium titanate anodes have been known since the 1980s. Li-titanate replaces the graphite in the anode of a typical lithium-ion battery and the material forms into a spinel structure. The cathode can be lithium manganese oxide or NMC. Li-titanate has a nominal cell voltage of 2.40 V, can be fast-charged, and delivers a high discharge current of 10 C, or 10 times the rated capacity. The cycle count is said to be higher than that of a regular Li-ion. Li-titanate is safe, has excellent low-temperature discharge characteristics, and obtains a capacity of 80% at -30°C (-22°F).

Discussion Questions:

1. Compare the materials used in the anode and cathode of Li–MnO₂ and Li-ion batteries. Predict the impact of battery efficiency and lifespan based on these materials.
2. Why are Li-ion batteries preferred for electric vehicles, while Li–MnO₂ batteries are used in medical devices and memory backup systems?
3. Why is a Li–MnO₂ battery considered non-rechargeable while a Li-ion battery is rechargeable?
4. Describe the function of the electrolyte and separator in lithium-based batteries. Why is a non-aqueous electrolyte preferred?

References:

- Jain and Jain, Engineering Chemistry, 16th Edition, Dhanpat Rai Publishing Company, New Delhi, 2013.
- P.H. Rieger, Electrochemistry, Second Edition (Reprint), Springer, Netherland, 2012.
- S. Vairam, Engineering Chemistry, John Wiley & Sons, 2014.
- Gary L Miessler Doland A. Tar, Inorganic Chemistry (Fourth Edition), Pearson, 2024.